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FORSKO
LEARN • GROW • SUCCEED
Syllabus
DATA STRUCTURES
ALGORITHMS
COMPUTER NETWORKS
DATABASE SYSTEMS

Major Course: Laboratory on Advanced Java Programming

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	**112224**	**Laboratory on Advanced Java Programming**	2	60	4Hrs.	50

Course Objectives:	1. To understand the concepts of advanced Java programming and object-oriented techniques. 2. To develop applications using exception handling and multithreading concepts in Java. 3. To understand GUI programming using AWT, Applets, layouts, controls, and event handling. 4. To design and implement interactive Java applications using AWT components and event-driven programming. 5. To understand database connectivity using JDBC and perform database operations such as create, insert, update, display, and delete.			
Course Outcomes:	On competition of the following syllabus the students will be able to - 1. Apply advanced Java concepts such as exception handling, multithreading, Applets, and event handling in application development. 2. Design and develop GUI-based applications using AWT controls, layouts, panels, buttons, and frames. 3. Implement event-driven programming techniques for solving mathematical and logical problems using Java. 4. Create interactive Java applications using Applets and AWT components. 5. Establish database connectivity using JDBC and perform CRUD (Create, Read, Update, Delete) operations on databases. 6. Develop Java applications integrated with relational databases for storing and retrieving data. 7. Debug, test, and execute Java programs effectively in different application environments.			
Contents		**Workload Allotted**	**Weightage of Marks Allotted**	**Incorporation of Pedagogies**
	List of Practical: 1. Write a program to show exception handling in java. 2. Write a java program for creating multiple catch blocks. 3. Write a program to show the concept of threads. 4. Write a program to show the concept of Applets. 5. Implements and create five buttons and put on different direction and centre by using border layout manager. 6. Write a program in AWT to create a frame with two buttons having text “Welcome” and “SGBAU” to respective command B1 and B2.			Use any pedagogical technique suitable for the topic

	7. Write a program to create a panel with two command buttons having different color to respective button as well as panel using AWT controls. 8. Write a program to calculate factorial of a number using AWT controls and events. 9. Write a program to calculate whether the given number is prime or not using AWT controls and events. 10. Write a program to calculate mathematical model using AWT controls and events. 11. Write a program in JDBC to create name of students table that contain the field student(Sid int, Sname varchar(20), marks int). 12. Write a program in JDBC to accept students data from student and register the new student in database. 13. Write a program in JDBC to display students information from students table. 14. Write a program in JDBC to receive marks from command line and update students marks or specified Sid. 15. Write a program in JDBC to receive marks from command line and delete the record of specified student with respect to the desirable marks.			
	Weblink to Equivalent MOOC on SWAYAM if relevant: 1. https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_cs110 2. https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_cs31 3. https://onlinedegree.swayam2.ac.in/mri22_01_d03_s1_cc3/preview 4. https://onlinedegree.swayam2.ac.in/mri23_07_d03_s4_cc02/preview 5. https://swayam.gov.in/			

Assessment Rubric: External Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI
PRACTICAL EXAMINATION
B.Sc. III, SEMESTER – VI (NEP)

Practical	**Course Code:** 112224	**Laboratory on Advanced Java Programming**	**Max Marks: 25**	**Time:** 4 Hrs.

Sr. No.	**Exercise**	**Marks**
1.	Single Program Writing, Algorithm, Flowchart, Execution, with Sample inputs and Outputs, and Printout	**10**
2.	Submitted Record book (Preferably handwritten)	**05**
3.	External Viva Voce	**10**

Assessment Rubric: Internal Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)		
Practical	**Course Code:** 112224	**Laboratory on Advanced Java Programming**	**Max Marks: 25**
Sr. No.	**Assessment Criteria**		**Marks**
1.	Attendance/Regularity in practical's / Students' overall performance		**05**
2.	Record book Preparation (Preferably handwritten)		**10**
3.	Internal Viva Voce		**10**